

Named Entity Recognition (NER)

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Agenda

- Motivation & Context
- Data Overview
- Baseline Model & Findings
- Enhanced Architecture
- Results Comparison
- Deployment Plan
- Future Directions

Motivation & Broader Context

→ Why NER?

- ◆ Extract structured facts (people, places, organizations) from unstructured text
- ◆ Power applications in search, question answering, compliance monitoring

→ Business impact

- ◆ Automate tagging to reduce manual review by 80%
- ◆ Enable real-time analytics on customer feedback, news streams

Data Overview

- **Source:** ner_dataset.csv (e.g., CoNLL-2003 style).
- **Size & Split:**
 - ◆ **47,959 sentences.**
 - ◆ **8 entity types:** **PER** (Person), **ORG** (Organization), **LOC** (Location), etc.
 - ◆ **Data split:** 60% Train, 20% Validation, 20% Test.
- **Label Distribution:**
 - ◆ **Common labels** like **PER**, **ORG**, **GPE**, **GEO**, and **TIM** have much **higher counts**, with **tens of thousands** of instances.
 - ◆ **Rare labels** such as **ART**, **EVE**, and **NAT** have **significantly fewer** instances, with some having **only a few hundred** instances.

Baseline Architecture

- **Model: BERT-base-cased** → token classification head + **CRF** (Conditional Random Field).
- **Training Details**
 - ◆ **Max length** = 128 tokens
 - ◆ **Batch size** = 32.
 - ◆ **Epochs** = 3.
 - ◆ **Loss: Class-weighted CrossEntropy loss** to handle imbalance.
- **Sequence-Level Constraints: CRF** layer to enforce valid **BIO tag sequences**.

Baseline Findings

→ Performance:

- ◆ **Val/Test micro-F1 $\approx$ 0.78.**
- ◆ **Strong performance on common tags: PER (0.90), GPE (0.92).**
- ◆ **Low precision and recall for rare classes like ART, EVE, and NAT:**
 - **ART: Precision = 0.47, Recall = 0.19, F1 = 0.27.**
 - **EVE: Precision = 0.40, Recall = 0.22, F1 = 0.28.**
 - **NAT: Precision = 0.32, Recall = 0.24, F1 = 0.27.**

→ Performance:

- ◆ **Label transition errors** (e.g., B-ORG to I-PER).
- ◆ **Context loss** on truncated sequences due to the max token limit (128 tokens).

Enhanced Model Design

Enhancement: Add **BiLSTM + CRF** on top of **BERT**.

- **BiLSTM:** Captures **longer-range token dependencies** by processing the sequence in both directions (past and future tokens).
- **CRF:** Ensures valid **BIO tag sequences** (e.g., no I-ORG without B-ORG).
- **Hyperparameters:**
 - ◆ **LSTM hidden size** = 256.
 - ◆ **Dropout** = 0.1 to reduce overfitting.
- **Why this enhancement?**
 - ◆ **BiLSTM** allows the model to capture better context over longer sequences.
 - ◆ **CRF** ensures better prediction coherence between consecutive tokens.

Enhanced Model Results

- **Test micro-F1** improved to **0.80**, an improvement of **+2 pp** over the baseline.
- **GPE/LOC recall** up by **~8%** (better performance on location-based entities).
- **Improved tag sequences:**
 - ◆ **More consistent BIO transitions:** The model avoids invalid tag sequences (e.g., B-ORG followed by I-PER).
- **Challenges:**
 - ◆ Still gaps in performance on extremely rare classes (e.g., ART, NAT)—requires more data to address these.

Unified Narrative

- **Problem:** Extract structured entities from **unstructured text**.
- **Baseline:** Strong on **common entity types** but weak on **rare entities** and **label transitions**.
- **Enhancement:** **BiLSTM + CRF** improves the model's ability to capture **sequence coherence**.
- **Outcome:**
 - ◆ **+2 pp** in micro-F1.
 - ◆ **Smoother tag sequences**.
 - ◆ **Better generalization on rare classes** but still gaps.

Deployment Plan

→ Model Serialization:

- ◆ Export model via **TorchScript** or **ONNX** for inference server deployment.

→ Inference API:

- ◆ Wrap model in a **FastAPI** endpoint.
 - **Input:** Raw sentence list.
 - **Output:** BIO-tagged tokens.

→ Scaling:

- ◆ Containerize with **Docker**.
- ◆ Deploy to **Kubernetes** for auto-scaling (CPU/GPU pods based on demand).

→ Monitoring & Logging:

- ◆ Track **per-entity F1** on incoming data.
- ◆ **Alert** if F1 drops by more than **5%**.

→ Why this deployment approach?

Kubernetes ensures efficient scaling, while **FastAPI** provides fast API response times for inference.

Roll-out Phases

- **Phase 1 (Staging):**
 - ◆ Batch inference on **historical data**.
 - ◆ Perform **manual spot-checks** and annotate errors.
- **Phase 2 (Pilot):**
 - ◆ **Real-time inference on 10% of production traffic**.
 - ◆ **Monitor latency and resource usage** for stability.
- **Phase 3 (Full Launch):**
 - ◆ **100% traffic** routed through the model.
 - ◆ Integrate **NER tags** into **downstream pipelines** for real-time processing.

Why this phased approach?

- **Staging** and **Pilot** allow for error correction before scaling to full production.

Future Directions

→ Data Augmentation:

- ◆ Use **back-translation** and **synonym swapping** to augment data for rare classes.

→ Domain-Adaptive Pretraining:

- ◆ Fine-tune the model on domain-specific **unlabeled corpora** to improve accuracy in specialized contexts.

→ Active Learning:

- ◆ Prioritize human labeling on uncertain predictions, especially for **rare entities**.

→ Model Compression:

- ◆ Use **DistilBERT** and **quantization** to optimize the model for **edge deployment** on low-resource devices.

Why these directions?

- **Data augmentation** and **domain adaptation** can help the model perform better on rare or domain-specific entities, while **model compression** is necessary for **edge deployment**.

Q&A

- Thank you!
- Questions?